DATA MANAGEMENT PLAN

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Research group	NFP (Nanostructured Films and Particles), INMA, University of Zaragoza
Programme	IberusINTERDOC / Campus Iberus, Horizon Europe MSCA-COFUND (G.A. 101178498)
Research line	CO_2 mitigation: capture, utilization and process decarbonization
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1. Purpose and scope

This plan describes how I collect, name, store, protect and share the data produced during my PhD. It follows the structure of the Horizon Europe data management plan (data summary, FAIR data, other research outputs, resources, security, ethics) and the Open Science obligations of the Grant Agreement, which require a regularly updated DMP, deposit in a trusted repository and open access to the data. It is a living document. I review it every six months and whenever my methods, materials or collaborators change.

The data come from my work on Microwave Swing Adsorption (MWSA), direct ink writing (DIW) of structured adsorbent monoliths, and the characterization of zeolite- and ceria-based materials. Some parts of this plan are already in place, and others start from now. Section 7 shows clearly the details.

2. Data summary

Most data are generated in the NFP laboratory or with the equipment of the University of Zaragoza. All data are new data produced by me. Literature data are used only for comparison and will always cited.

Data type	Raw format	Software / instrument	Purpose
CO_2 adsorption–desorption (microwave system)	.txt, .csv	Microwave adsorption–desorption set-up	Performance of the Zeolite Powder, monoliths during adsorption and microwave regeneration
Mass spectrometry (MS) and microwave data	.txt, .csv	Quadera 32 and the microwave system software	Gas compositions and record of analysis and to check the adsorption capacity of CO_2.
Thermogravimetric analysis (TGA)	.xlsx, .csv	TGA instrument	Thermal stability of the Zeolite pastes and to check the loss of water content and other volatile components.
BET surface area and porosity	.txt	Physisorption analyser	Surface area and pore structure for the silica gel
X-ray computed tomography (XCT)	.zinspect	XCT scanner software	Print quality, porosity and internal channels
Rheology of the inks	.rwd, .xlsx	Rheometer software	Viscosity and shear behaviour of the pastes
3D printing files	.stl, .gcode	Solid Edge, Cura, Sygnis, processed G-code	Design and printing of the monoliths
Infrared camera videos	.ravi	IR camera	Temperature distribution during microwave adsorption–desorption
Sample register.	.xlsx	Excel	Record of each monolith, weight and dimensions before and after calcination
Processed data and documents	.xlsx, .opju, .docx, .pptx	Excel, Origin, Word, PowerPoint	Analysis, figures, reports and presentations

3. FAIR data

3.1 Findable

Every monolith has a unique code (Section 4), and every file name contains that code, so any result can be traced back to the exact sample. Datasets will be published in Zenodo receive a persistent DOI. I will also link my ORCID to my repository records and add keywords, a description and the funding acknowledgement (MSCA-COFUND, G.A. 101178498) to each record.

3.2 Accessible

Selected data will be deposited in Zenodo and in Zaguan (the institutional repository of the University of Zaragoza). Both are free and open to everyone, and both keep the data for the long term. Data are shared under one of three access levels (Section 6). Metadata will always be public, even when the data themselves are restricted.

3.3 Interoperable

Proprietary files (.rwd, .zinspect, .ravi, .xlsx) are kept as raw data, and I will also export a copy in an open format (CSV or TXT for tables, TIFF or PNG for images, MP4 for videos) for sharing. Standard SI units and clear column headings are used in all tables and figures.

3.4 Reusable

Each dataset includes a README file with the method, the instrument settings, the units and the software needed. Experiments will be proceeded on three copies of each monolith (a, b and c), which lets me check repeatability. Open data will use the latest version of CC BY (currently 4.0) or CC0, so others can reuse them if they give credit where required.

4. Sample naming

4.1 Sample code

Every monolith has a code made of five parts, written in the same order every time. The date is the date on which the monolith was made. For example:

ZZ 10 B6 Zeo a (20-09-2026)

Part	Meaning	Example
ZZ	Geometry of the monolith	ZZ = zigzag
10	Infill pattern of the pores	Values used so far: 10, 18, 25 and 35
B6	Paste batch number	B6 = batch 6
Zeo	Material type	Zeo = zeolite (short code to identify material)
a, b, c	Copy of the same monolith	Three copies of each type, used to check repeatability

4.2 Current sample set (batch B6, zeolite)

Batch B6 contains four infill patterns with three copies each, so twelve monoliths in total. The date of each monolith is written in the excel file name as sample register.

Copy a	Copy b	Copy c	Infill pattern
ZZ 10 B6 Zeo a (date)	ZZ 10 B6 Zeo b (date)	ZZ 10 B6 Zeo c (date)	10
ZZ 18 B6 Zeo a (date)	ZZ 18 B6 Zeo b (date)	ZZ 18 B6 Zeo c (date)	18
ZZ 25 B6 Zeo a (date)	ZZ 25 B6 Zeo b (date)	ZZ 25 B6 Zeo c (date)	25
ZZ 35 B6 Zeo a (date)	ZZ 35 B6 Zeo b (date)	ZZ 35 B6 Zeo c (date)	35

4.3 Sample register

All monoliths are recorded in one Excel sample register. Each row is one monolith and includes the full code, the date, the geometry, the infill pattern, the batch, the material, the copy (a, b or c) and a notes column for printing and calcination conditions. Every new monolith is added to the register on the day it is made. The register will be store in personal computer and shared OneDrive.

4.4 File names

Next file names will use underscores instead of spaces, so they work on every system. They start with the sample code, then the technique, the measurement date (year-month-day) and a version number:

ZZ_10_B6_Zeo_a_TGA_20260920_v01.xlsx

Raw files will not be edited or overwritten. Processed files get a new version number (v01, v02, …), so the history of the analysis is always clear. Files that are not linked to one sample, such as rheology of a paste, use the batch code instead (for example B6_Zeo_Rheology_20260915_v01.xlsx).

5. Folder structure and documentation

Folder	Content
01_Raw_Data	Untouched instrument files, with one subfolder per technique (MW_ads_des, MS, TGA, BET, XCT, Rheology, IR_videos)
02_Processed_Data	Excel and Origin files with calculated results
03_3D_Printing	STL files, Solid Edge designs, Cura projects and G-code
04_Figures	Final graphs and images for papers and presentations
05_Documents	Reports, drafts, presentations and meeting notes
06_Metadata	README files, sample register and experiment log

To document my work, I will:

- keep an experiment log for every experiment (date, sample code, conditions, settings and observations);
- add a README file to every dataset that explains the files, units and column names;
- record printing parameters, paste composition and calcination conditions in the sample register;
- write down any problem, such as a failed print or a sensor issue, so that the data are interpreted correctly later.

6. Open science and data sharing

6.1 Principle

I follow the principle “as open as possible, as closed as necessary”. As required by the Horizon Europe Grant Agreement, I will deposit my research data in a trusted repository as soon as possible and within the deadlines given in this plan. Through the repository, the data will be openly available under the latest version of the Creative Commons Attribution licence (CC BY, currently 4.0) or the Creative Commons Public Domain Dedication (CC0). Data are kept closed only when opening them would go against legitimate interests, such as commercial exploitation, or against other constraints of the Grant Agreement. In that case the reason is written in this plan.

6.2 Repositories

- **Zenodo:** the main trusted repository for datasets. Each record receives a DOI and public metadata.
- **Zaguan (University of Zaragoza):** institutional repository for my articles, my thesis and related data, in agreement with the university's open access policy.

6.3 Access level of each dataset

Level	Data	Licence and justification
Open	Data that support published results: processed data, final figures, selected raw data, final STL files and print settings.	CC BY 4.0 (or CC0 for simple numerical data). No justification needed.
Restricted (temporary)	Data of ongoing work, shared only with my supervisor and project partners.	Closed only until the related article is published. The reason is that early release could affect publication and my results have not been checked yet.
Closed (confidential)	New ink and paste formulations and results that may be protected, for example by a patent.	Closed to protect legitimate commercial interests. The situation is reviewed at least once a year, and the data are opened as soon as the reason no longer applies.

My supervisor and I will decide the level of each dataset together. The metadata of every record will always public, even if the data are restricted or closed, so that the dataset can be found. When needed, I will use an embargo in Zenodo so that data become public only after the related article is published.

6.4 Deposit deadlines

Situation	Deadline
Data behind a journal article or conference paper	Deposited at the latest on the day of publication and cited in the article with its DOI.
Data behind the thesis	Deposited before the thesis defence.
Other useful data (for example complete experimental campaigns)	Checked and deposited at least once a year, during the annual update of this plan.
Closed data	Access reviewed once a year and opened as soon as the reason for closing no longer applies.

6.5 Information needed to reuse and validate the data

For each deposited dataset I will add, in the repository record, the information that others need to reuse or validate it: a README file, the sample register, the experiment protocol, and the name and version of the software and instruments used. Where the software is commercial, I also provide the data in an open format.

Tool	Used for	What I provide
Solid Edge, Cura, Sygnis	Design and printing of the monoliths	STL files, printing settings and the processed G-code
Quadera 32 and the microwave system software	MS and microwave records	Exported CSV or TXT files and the software version
Rheometer software	Ink rheology (.rwd)	Exported Excel or CSV tables of viscosity and shear rate
XCT software	Tomography (.zinspect)	Exported image stack (TIFF) and the scan parameters
IR camera software	Thermal videos (.ravi)	MP4 copy and the camera settings
Origin and Excel	Data analysis and figures	Processed tables in CSV format and a short description of the calculations

7. Storage, backup and security

Data are stored at three levels: my laptop for daily work, the University of Zaragoza OneDrive as the main backup, and the repositories for long-term preservation. The table below shows the status and my plan.

Task	Status	Details
Laptop working copy	In place	All current data are on my laptop, which is password protected and updated regularly.
University OneDrive	Started with my supervisor	Part of the data is already uploaded. The rest will be moved to the folder shared with Prof. Reyes Mallada.
Monthly OneDrive upload	From now	New data are added every month, and always after an important experiment.
Zenodo account	From now	Account created and linked to my ORCID. The first datasets will be uploaded with the correct access level.
Zaguan repository	Planned	Used for published articles and, at the end, the thesis and related data.

8. Other research outputs

Besides data files, my research produces other outputs that will also be managed and, when possible, shared: 3D designs (STL) and printing settings, ink and paste recipes, experimental protocols, SOPs for the instruments that I used in my research, analysis files (Origin), Trainings, articles, conference presentations and the final thesis. Articles will be published in open access journals, in line with Horizon Europe rules.

9. Resources, ethics and responsibilities

Resources. Data management does not need extra funding. OneDrive is provided by the university, and Zenodo and Zaguan are free. The time needed for documentation is part of my normal research work.

Ethics and IPR. This research does not involve human public participants or personal data, so General Data Protection Regulation (GDPR) does not apply to it. Intellectual property is handled according to the rules of the University of Zaragoza, the IberusINTERDOC programme and the agreements with partners. Before any dataset is made public, I will check with my supervisor that it contains nothing that needs protection.

Person	Responsibility
Shayan Gul	Collects, names, documents, backs up and uploads the data; keeps the sample register; updates this plan.
Prof. Reyes Mallada	Supervises data management and approves what is shared openly or kept confidential.
Collaborators (UPV, others)	Follow the same rules for shared data and give proper credit in publications.

10. Action plan

I will,

- Finish uploading all existing data to the shared OneDrive folder, using the folder structure in Section 5.
- Add every monolith to the sample register and apply the naming rules to all new files.
- Write a README file and an experiment log entry for each technique.
- Upload the first datasets to Zenodo, and set open, restricted or confidential access for each one.
- Add new data to OneDrive every month.
- Deposit published articles and, at the end, the thesis in Zaguan.
- Review and update this plan every six months and before each publication.